Wrist unit wiring reference (PROPOSED, NOT BUILT, NOT TESTED)

Only compile, boot and the camera-probe failure were run, on a Wokwi XIAO ESP32-S3 with OLED, button and LED (Wokwi has no camera part).

Power (proposed)
USB-C 5 V from a power bank
3V3 regulator on the XIAO feeds ESP32-S3, camera and OLED;
5 V pin feeds the LED stage
Battery operation not designed

NOT TESTED

Seeed XIAO ESP32S3 Sense (ESP32-S3R8, 8 MB PSRAM)

Camera on the Sense expansion board
proposed OV5640 AF (vendor: OV5640 or OV3660 may ship)
board-to-board connector: NO external wiring
XCLK 10 PCLK 13 VSYNC 38 HREF 47
Data 15, 17, 18, 16, 14, 12, 11, 48   SCCB SDA 40 / SCL 39

NOT TESTED

Camera, microphone and microSD GPIOs are used by the board.
D8-D10 share the SD pins and are left unused.

Firmware (about 80 lines)
button -> LED on -> JPEG frame -> HTTP POST
reply (3 lines) -> OLED
no OCR / ML on this unit

NOT TESTED

D4 SDA
D5 SCL

SSD1306 OLED 128x64, I2C
VCC 3V3 | GND | SDA | SCL

SIMULATED

D1

CAPTURE push-button
D1 / GPIO2 to GND
(internal pull-up)

SIMULATED

D0

White illumination LED stage
D0/GPIO1 -> 1 k -> NPN base
LED + series resistor from 5 V to collector; emitter to GND
(resistor from the LED datasheet)

PROPOSED

Receiver tested with Python clients; the firmware side of the POST has not run.

Link to the Python companion

Wi-Fi 2.4 GHz, same network as the laptop → **One HTTP POST: image/jpeg body to /api/wearable/upload; 3-line text reply**

TESTED